

PolyAgora V7.6 — First-Cut Findings

Date: 2026-05-14 **Status:** Pre-dashboard spike. Four variants tested (v76 α , v76 β , v76 γ , v76 γ -full). Architecture validated; quantitative alpha over naive diversification is regime-dependent and not closed by the spec's full MRTP composite under cheap implementations. **Spec source:** [docs/PolyAgora_Meta_Regime_Steering_Layer_V0.1.pdf](#), [docs/ChatGPT - PolyAgora - Fixed Star Fund - FSF.pdf](#), [docs/PolyAgora.pdf](#) (all dated 2026-05-14, same content in three formats).

Executive summary

The minimum-viable meta-regime steering layer **delivers what the spec describes** — better Sharpe than any single manifold, strong "transition maneuverability" in 2008 / COVID, and robustness across hyperparameter choices. **But** the steering's marginal alpha over a naive equal-weight blend of the same manifolds is concentrated in the 2008–2016 half of the sample, and is neutral-to-negative in 2017–2026. We tested four progressively more sophisticated meta-layer formulations: rolling-Sharpe-only (v76 α), regime-gated admissibility (v76 β), full MRTP composite with persistence + adverse-CVaR (v76 γ), and MRTP composite + regime-gate (v76 γ -full). **None close the 2017–2026 gap; the more complex variants make it worse.** Under cheap implementations the spec's full MRTP composite is not predictive — its F_i (persistence autocorr) and D_i (regime-mismatch CVaR) terms are forward-looking only in *interpretation*, while their data inputs remain rolling backward signals that lag regime changes as much as rolling Sharpe does.

Headline (2008-04 → 2026-04, 4,669 trading days):

Series	Sharpe	Sortino	Calmar	MaxDD	CAGR
v74d_q (M1)	0.47	0.56	0.18	-9.7%	1.79%
momentum_12_1 (M2)	0.52	0.67	0.22	-14.7%	3.26%
defensive (M3)	0.62	0.93	0.18	-19.0%	3.51%
cash (M4)	—	—	—	0.0%	0.00%
naive 1/4 blend	0.74	1.00	0.33	-6.6%	2.21%
v76α (rolling-Sharpe gate)	0.82	1.08	0.32	-10.3%	3.23%
v76β (MRTP-lite, regime-gated)	0.80	1.03	0.31	-12.4%	3.85%
v76γ (full MRTP: S + F – D, A = 1)	0.72	0.92	0.29	-10.5%	3.10%
v76γ-full (full MRTP + regime A)	0.73	0.94	0.30	-10.5%	3.19%

What's new in V7.6

V7.6 is a **structural shift**, not another single-engine refinement. The prior V7.x line iterated one alpha engine (V7.3 / V7.4 / V7.5) inside the same Reference Polygon. V7.6 introduces a **two-level architecture per spec §2** — strategy manifolds at level 1, a meta-regime steering layer at level 2 — and treats the prior engines as one of several manifolds rather than as the system itself. Concretely new in V7.6:

1. **Four manifolds instead of one engine.** v74d_q (the V7-line winner) is now M1 of four. M2 is plain 12-1 momentum. M3 is a deliberately simple frozen defensive mix (40% TN + 20% FGBL + 20% GC + 20% DX). M4 is pure CASH — the absorbing rest-state for when no manifold has positive realized Sharpe. Each manifold is frozen during meta-layer validation per spec §12.A.

2. **Meta-regime steering layer.** A slow softmax-over-manifolds with admissibility gating, computed once per close from `realized ≤ t-1`. Four variants tested, progressively more sophisticated: v76α (Sharpe-only), v76β (regime-conditioned admissibility), v76γ (full MRTP composite $\alpha \cdot S + \beta \cdot F - \gamma \cdot D$), v76γ-full (MRTP + regime A).
3. **Smooth probabilistic steering, no binary regime switches.** EWM halflife 10 days plus a 15% allocation floor (v76α) gives 0 days with $|\Delta W_i| > 5\text{pp}$ over 4,669 trading days. Spec §4's "slower / coarser / lower entropy" mandate is satisfied.
4. **Explicit recoverability framing.** The product optimization target is no longer "beat the benchmark." It is the spec §1 objective set — absolute returns, recoverability, maneuverability, low correlation to traditional asset classes, institutional robustness.
5. **A negative result we are publishing.** v76γ and v76γ-full implement the spec's full MRTP composite from cheap inputs derivable from existing v75 diagnostics. They do not close the recent-period gap to naive 1/4 diversification — they widen it. Documented so the next iteration does not retrace this path.
6. **Empirical reference line: naive 1/4.** A fixed equal-weight blend of the four manifolds is now the reference any v76 variant must clear. Its full-sample Sharpe of 0.74 is the diversification-only baseline; the meta-steering layer's marginal value is whatever sits above that line.

Nothing in the V7.4 / V7.5 line is deprecated. v74d_q remains the V7-line winner, and is the M1 manifold in V7.6.

Architecture

Two-level construction per spec §2:

Level 1 — Strategy manifolds (frozen, no further tuning):

- **M1 = v74d_q** — the PolyAgora V7 manifold (`make_v75_q_signal` with `α_M=0`, `include_mom_eigen=False`, `K=2`, `τ=2`, `λ=0.5`, VAIDM × ADD Q polygons). Maps to spec §3.2: transitions, fragmentation, boundary instability.
- **M2 = momentum_12_1** — the 12-1 momentum baseline from `polyagora_v63_partner_engine`. Maps to spec §3.1: persistent trends, coherent low-vol expansions.
- **M3 = defensive** (new, `polyagora_v76_engine.defensive_signal`) — frozen 40% TN + 20% FGBL + 20% GC + 20% DX. Maps to spec §3.3: rupture, collapse geometry.
- **M4 = cash** (new, `polyagora_v76_engine.cash_signal`) — zero real-asset weights. The absorbing rest-state for when no manifold has positive rolling Sharpe.

Level 2 — Meta steering layer: softmax-of-rolling-Sharpe with two variants of the admissibility gate.

```
v76α: W_i(t) = exp(λ·MRTP_i(t)) / Σ_j exp(λ·MRTP_j(t))
v76β: W_i(t) = A_i(t) · exp(λ·MRTP_i(t)) / Σ_j A_j(t) · exp(λ·MRTP_j(t))

where
MRTP_i(t) = annualized rolling Sharpe of manifold i's returns over [t-63, t-1]
A_i(t)     ∈ [0.10, 1.00] — see "Admissibility (v76β)" below
EWM smoothing on W_i with halflife 10 days, then min_floor 0.15 (v76α only).
```

Anti-hindsight: `MRTP_i(t)` reads `returns_i.shift(1)`; v75 diagnostics at `t` already depend only on `realized ≤ t-1`, so they pair with `manifold_weights[t]` without further shift.

Full MRTP composite (v76γ, v76γ-full)

Spec §8: $\text{MRTP}_i = \alpha \cdot S_i + \beta \cdot F_i - \gamma \cdot D_i$. Anti-hindsight is preserved by computing every term from data $\leq t-1$ — "forward-looking" in the spec terminology means *projects expected future state from a model fit on the past, not reads the future*.

```

S_i(t) = tanh( rolling Sharpe of returns_i over [t-63, t-1] / 2 ) # → [-1, 1]
F_i(t) = lag-21 autocorrelation of a regime coord over [t-252, t-1]:
    v74d_q ← autocorr(dQ_τ) - transition activity persistence
    momentum ← autocorr(T_τ) - trend persistence
    defensive ← autocorr(V_τ) - vol-stress persistence
    cash ← -max(S_i for i ≠ cash) - option value when nothing works
D_i(t) = tanh( 50 · CVaR_5%( returns_i_τ | sign(V_τ, T_τ) ≠ sign(V_t, T_t) ) ) # → [0, 1]

```

v76γ runs the composite with $A_i \equiv 1$ (no gate — MRTP does the regime work); **v76γ-full** keeps the v76β regime-gated A_i . Defaults $\alpha = \beta = 1$, $\gamma = 0.5$, lookbacks 63/252/252, EWM halflife 10d, floor 0.05.

Anti-hindsight is enforced at three points: (1) `_rolling_sharpe` applies `.shift(1)` before windowing; (2) `_rolling_autocorr` does the same on the regime coord; (3) `_conditional_cvar` uses `bin_V = sign(V.shift(1))` and indexes returns via `r.shift(1)`. v75 diagnostics at row t already depend only on `realized ≤ t-1` by the v75 invariants, so the diag-side inputs need no further shift.

Admissibility (v76β, MRTP-lite)

Each manifold's regime-conditional admissibility is computed directly from v75's existing diagnostics output — no new feature stack:

```

A_v74d_q = σ(2.0 · [1.5·dQ+ + 1.0·(d_RP-0.4)+] - 0.5)
A_momentum = σ(3.0 · [|T| - 1.5·dQ+] - 0.5)
A_defensive = σ(3.0 · V)
A_cash = σ(-3.0 · max(rolling_sharpes, 0))

```

then floored at 0.10. The mapping echoes the spec's conceptual interpretation (§3): V7 wins under transition activity (`dQ`, `d_RP`); momentum wins under stable trends (high `|T|`, low `dQ`); defensive wins under VIX stress (high `V`); cash wins only when no manifold's recent realized Sharpe is positive.

Results

Robustness — sensitivity sweep (500 configs)

Grid over `lookback ∈ {21, 42, 63, 126, 252}`, $\lambda \in \{0.5, 1, 2, 4, 8\}$, `halflife ∈ {1, 5, 10, 21, 60}`, `floor ∈ {0, 0.05, 0.10, 0.20}`.

- 4-manifold v76α Sharpe distribution: min 0.58, 25th 0.73, median 0.76, 75th 0.79, max 0.86.
- Every config beats every single static manifold (best static = defensive at 0.62).
- Median Sharpe rises monotonically with `floor`: 0.75 at floor=0 → 0.78 at floor=0.20.
- Lookback=63 is the median winner (0.81); lookback=252 collapses (0.68).
- λ is approximately flat between 0.5 and 8.

Conclusion: the chosen (`lookback=63`, $\lambda=2$, `halflife=10`, `floor=0.15`) lands near the 90th percentile of the grid. Sharpe 0.82 is not a hyperparameter accident.

Crisis windows — the spec's "transition maneuverability"

Window	naive	v76 α	v76 β	v76 γ	v76 γ -full
2008 GFC (Apr08–Jun09)	-0.81	+0.55	+0.94	+0.76	+0.79
COVID (Feb20–Dec20)	+0.54	+1.37	+1.14	+0.44	+0.31
2022 rates (Jan22–Dec22)	-0.49	-0.71	-0.89	-0.48	-0.55
Last 3 years (Jan23–)	+0.85	+0.72	+0.65	+0.49	+0.55

All four steered variants substantially outperform naive in 2008. In COVID v76 α is the clear winner — adding F_i and D_i (γ , γ -full) *halves* the COVID Sharpe (1.37 $\rightarrow$ 0.44). v76 γ is the only variant to beat v76 α on 2022 rates, but at the cost of much worse COVID and trend-period results. **No variant beats naive in 2022 rates or the recent 3-year trend.**

Temporal split — the headline finding

Sharpe in each window:

Window	naive	v76 α	v76 β	v76 γ	v76 γ -full
2008-04 $\rightarrow$ 2016-12 (first half)	0.78	0.99	1.03	0.93	0.94
2017-01 $\rightarrow$ 2026-04 (second half)	0.70	0.65	0.56	0.52	0.53
2008-04 $\rightarrow$ 2019-12 (pre-COVID)	0.81	0.95	0.97	0.89	0.92
2020-01 $\rightarrow$ 2026-04 (COVID-on)	0.60	0.56	0.43	0.35	0.33

MaxDD in each window:

Window	naive	v76 α	v76 β	v76 γ	v76 γ -full
2008-04 $\rightarrow$ 2016-12	-5.1%	-6.1%	-7.4%	-6.9%	-6.9%
2017-01 $\rightarrow$ 2026-04	-6.6%	-10.2%	-12.4%	-10.5%	-10.5%
2008-04 $\rightarrow$ 2019-12	-6.0%	-8.6%	-10.9%	-10.5%	-10.4%
2020-01 $\rightarrow$ 2026-04	-6.6%	-10.2%	-12.4%	-10.0%	-10.5%

The entire v76 alpha over naive comes from the first half of the sample. In 2017-2026 every steered variant underperforms naive on Sharpe and on MaxDD. **More machinery does not help — Sharpe in 2017-2026 decreases monotonically with sophistication:** α (0.65) $>$ β (0.56) $>$ γ -full (0.53) $>$ γ (0.52). The 2020-2026 window is starker still: naive 0.60 vs γ 0.35. Adding the spec's F_i and D_i terms makes the trend-period drag *worse*, not better.

Year-by-year vs naive (full sample, 19 years):

```

 $\alpha$  wins over naive: 13/19 years, mean +1.04pp/yr
 $\beta$  wins over naive: 13/19 years, mean +1.69pp/yr
 $\beta$  wins over  $\alpha$ :      13/19 years, mean +0.65pp/yr

```

β has higher mean alpha than α (full sample) — but pays for it in worse drawdowns (2022 -2.4pp vs α 's -1.1pp; 2023 -4.4pp vs α 's -3.2pp). γ and γ -full underperform both α and β across recent years. The pattern is unambiguous: the steering layer earns its keep in regime transitions and gives some back in persistent trends, and adding more terms to the meta-score does not recover the trend-period drag — it amplifies it.

Manifold allocation behavior

v76 α (Sharpe-only gate, with 15% floor):

```
v74d_q:      mean 0.285  std 0.205  min 0.104  max 0.688
momentum_12_1: mean 0.248  std 0.172  min 0.104  max 0.688
defensive:    mean 0.306  std 0.217  min 0.104  max 0.689
cash:         mean 0.160  std 0.098  min 0.104  max 0.657
days with |ΔW_i|>5pp: 0 across all manifolds
```

v76β (regime gate, no W-floor since A_i provides natural floor):

```
v74d_q:      mean 0.317  std 0.316  min 0.000  max 0.995
momentum_12_1: mean 0.264  std 0.264  min 0.000  max 0.992
defensive:    mean 0.335  std 0.340  min 0.000  max 0.999
cash:         mean 0.084  std 0.144  min 0.000  max 0.849
days with |ΔW_i|>5pp: v74d_q 6, mom 9, def 11, cash 1
```

v76β is materially more concentrated — three manifolds reach >99% allocation at some point. This explains both its higher absolute returns in transitions and its worse drawdowns when its regime call is wrong. v76α stays gently smoothed.

Regime calls (W_i snapshots, v76α with 15% floor):

```
2008-09 GFC peak:      def 0.91  mom 0.05  v74d 0.05  ✓
2020-03 COVID:        def 0.90  mom 0.05  v74d 0.05  ✓
2020-12 rally:         v74d 0.91  mom 0.05  def 0.05  ✓
2022-06 rates:         mom 0.81  v74d 0.14  def 0.05  ✓
2024-12 trending:      mom 0.88  v74d 0.05  def 0.07  ✓
```

Regime detection is qualitatively right in all five test points.

Honest assessment

The v76 architecture works as the spec describes. Four-manifold composition is clean, regime detection is qualitatively correct, sweep robustness is strong, and crisis-window performance is materially better than any single manifold or naive diversification. The spec's "transition maneuverability" promise is empirically delivered.

But the steering's quantitative alpha over naive diversification is regime-dependent and fragile. A fixed equal-weight blend of the four manifolds achieves Sharpe 0.74 with -6.6% MaxDD. v76α adds Sharpe (+0.08) and CAGR (+102bps) over the naive blend in full-sample terms, **but the wins are concentrated in 2008-2016 and the losses are concentrated in 2017-2026**. In the recent 6 years, naive 1/4 beats every v76 variant on Sharpe and MaxDD.

MRTP-lite (v76β) does not close this gap. Replacing the trivial admissibility gate with regime-conditioned A_i from v75's already-computed (T, V, d_{RP}, dQ) coords *amplifies* both ends of the trade — it concentrates harder when its call is strong (Sharpe 0.94 in GFC vs 0.55 for v76α) and concentrates harder when its call is wrong (2022 rates: -0.89 vs -0.71). Net Sharpe drops slightly (0.80 vs 0.82); net CAGR rises (3.85% vs 3.23%); net MaxDD worsens (-12.4% vs -10.3%).

Full MRTP (v76γ, v76γ-full) does not close it either — it makes the recent-period gap worse. This is the substantive negative finding of this iteration. The spec's full composite $M RTP_i = \alpha \cdot S_i + \beta \cdot F_i - \gamma \cdot D_i$ was supposed to overcome the lagged-Sharpe limitation by adding "future maneuverability" (F_i) and "expected damage if wrong" (D_i). Under the cheapest defensible implementation — lag-21 autocorrelation of regime coords for F , regime-bin-mismatch CVaR for D — neither term is actually predictive. Specifically:

- $F_i = \text{autocorr}(\text{regime_coord})$ over 252d tells us *how persistent the regime has been*, which is mathematically very close to *what already happened*. When a long trend is about to reverse, autocorr is at its peak right when the signal is most wrong.

- $D_i = \text{CVaR}_5\%(\text{returns} \mid \text{sign-mismatched regime bin})$ over 252d puts a static-historical loss prior on each manifold. It updates slowly and doesn't actually predict damage on the specific upcoming regime.
- Subtracting D from MRTP then penalizes whichever manifold has had ugly tails on regime mismatches — but the manifold with the worst historical mismatch-tail is often defensive (it gets crushed in rates-up regimes), so D ends up disproportionately suppressing defensive *at exactly the moment we need it most*.

Empirically: $v76\gamma$ underperforms $v76\alpha$ in every temporal window (Sharpe $0.65 \rightarrow 0.52$ in 2017-2026, $0.56 \rightarrow 0.35$ in 2020-2026). The COVID Sharpe collapses from 1.37 ($v76\alpha$) to 0.44 ($v76\gamma$). The 2022-rates Sharpe improves modestly ($-0.71 \rightarrow -0.48$) but the net effect across regimes is unambiguously worse.

Forward-looking $\neq$ predictive. Anti-hindsight is preserved in every term — F and D read only data $\leq t-1$. But "doesn't peek at the future" and "predicts the future" are different properties. F and D as cheaply implemented are forward-looking only in *interpretation*; their *information content* is essentially the same backward-window rolling statistics as S . Adding redundant lagged signals to a meta-score doesn't reduce its lag.

The structural bottleneck is the regime model, not the score composition. All four variants assume the meta-state can be summarized by (V, T, d_{RP}, dQ) plus rolling statistics of these. None of these are leading indicators — they are reactive to realized price and volatility. To genuinely move the needle on 2017-2026 we would need either:

- External leading signals — VIX term structure slope, yield curve shape, futures positioning, credit spreads — that price *expected* future regime change rather than measure realized regime state.
- A regime-prediction model (HMM, Markov chain, supervised classifier) trained walk-forward on past regime transitions to actually forecast $P(\text{regime}_{\{t+k\}} \mid \text{history}_{\{\leq t\}})$. Substantial engineering, possible overfit risk.
- Acceptance that $v76$ is structurally a crisis-defense overlay and stop building meta-machinery on top.

Sharpe > 1.5 target is out of reach with this approach. The best 4-manifold $v76\alpha$ config in the sweep peaks at Sharpe 0.86 over full sample, ~ 0.60 over the most recent 6 years. Reaching the spec's institutional bar would require either: (a) materially better manifolds (cleaner momentum, less drawdown-prone defensive), (b) genuinely predictive leading signals (external data feeds or a regime-prediction model), or (c) cash floor / volatility targeting overlays that aren't part of the meta-steering spec.

Path forward

The $v76\gamma$ spike answered the open question from the previous iteration: **the spec's full MRTP composite, under any cheap implementation derivable from existing $v75$ diagnostics, is not predictive enough to close the 2017-2026 gap.** This collapses the prior decision tree to three concrete options.

Option 1 — Ship $v76\alpha$ as a crisis-defense overlay. Treat the 4-manifold construction with the trivial Sharpe gate as the production default. Frame it honestly: diversification + steering = small Sharpe lift over naive, neutral-to-negative recent-period drag, but materially better crisis defense (Sharpe $0.55-1.37$ vs $-0.81-0.54$ for naive in 2008/COVID). The 15% W-floor keeps drawdowns bounded. Build the dashboard around this — manifold-allocation panel, meta-state strip, three-system comparison per spec §12.D.

Option 2 — Ship $v76\beta$ as the higher-return / higher-risk variant. Use the regime-gated version for clients who want more aggressive transition-following (+62bps CAGR vs $v76\alpha$ at the cost of -2.1pp on MaxDD). Pair with explicit drawdown warnings. Lower priority than Option 1 — the trade is not obviously a Pareto improvement.

Option 3 — Stop building meta-steering machinery on top of internal diagnostics. v76 γ established the ceiling of what's reachable from (V, T, d_{RP}, dQ) plus rolling statistics of these. To exceed it, the next spike must introduce *external leading information* — VIX term structure, yield curve, credit spreads, futures positioning — or a real regime-prediction model trained walk-forward. Both are substantial engineering, with non-trivial overfit risk. Defer until Option 1 is shipped and we know whether the crisis-defense framing is what clients actually want.

Explicitly rejected paths after this iteration:

- Further variants of the MRTTP composite (different weightings of S/F/D, different lookbacks for F/D, different regime-bin definitions) — the structural problem is the information content of the inputs, not the composition.
- More elaborate admissibility gates derived from v75 diagnostics — v76 β already showed this saturates.
- Online learning or parameter-fitting against realized returns — guard-rail per spec §16.

Recommendation: Option 1. Build the dashboard around v76 α with v76 β as an alternate. Frame the product honestly as a crisis-defense + diversification overlay, not as an institutional Sharpe-maximizer. If subsequent work wants to push past Sharpe ≈ 0.82 , the substantive next bet is external leading data feeds (Option 3), not more iterations on the internal-diagnostics meta-score.

What NOT to do (spec §16 guard-rails)

These guard-rails held in this iteration and should hold in any next step:

- No further tuning of v74d_q parameters. M1 is frozen.
- No online learning, no parameter-fitting against forward returns, no meta-meta.
- No high-frequency switching. v76 α achieves 0 days with $|\Delta W_i| > 5pp$; v76 β has 6–11. Either is well within "slow/coarse/low-entropy."
- No carry / convexity / macro overlays on top of v76 until either Option 3 closes the temporal-split gap or we've explicitly accepted v76 as a crisis-defense product.

Files

- `polyagora_v76_engine.py` — engine. `defensive_signal`, `cash_signal`, `MetaConfig`, `meta_blend` (v76 α), `compute_regime_admissibility`, `meta_blend_mrtt` (v76 β), `MrttConfig`, `compute_mrtt_score`, `meta_blend_gamma` (v76 γ).
- `run_v76_check.py` — runs all four manifolds + naive + v76 $\alpha/\beta/\gamma/\gamma$ -full, prints full comparison incl. temporal split. CLI flags: `--lookback`, `--lam`, `--halflife`, `--floor`, `--no-cash`.
- `run_v76_sweep.py` — 500-config sensitivity sweep over the four meta hyperparameters.
- `polyagora_v76_outputs/` — weights, returns, equity, diagnostics, manifold allocations, admissibility frames, MRTTP scores for all seven series.